

WC
6/11

Week 7

* Meeting Cancelled as not required *

Log

7/11 Read Chapter 11 'The fundamental group' in Lickorish book

7/11 Read chapter 8.3 'The fundamental group' in Crossley Book

9/11 Read chapter 2 'The fundamental group' in Crowell, Fox book

10/11 The fundamental group is a mathematical group associated to any given pointed topological space that provides a way to determine when two paths can be continuously deformed into each other

11/11 The knob group is the fundamental group of an orientated link

Supervisor Signature: Anttho

WC
13/11

Week 8

* Meeting cancelled and heavily reduced work *
due to medical reasons

Log

13/11 cancelled meeting with Anita

17/11 Feedback from first logbook:

- Increase mathematical content
- Show worked examples

Supervisor Signature 

WC
20/11

Week 9

* Meeting cancelled and no log due to *
medical reasons

Log

21/11 Cancelled Meeting with Anitha

Supervisor Signature: Anitha

WC
27/11

Week 10

Meeting 30/11

Discussed what was expected from progress report and expressed how I would be limited due to previous two weeks.

Log

29/11 Read chapter 2 in 'An introduction to knot theory and the knot group' by Larson Lhovi.

- Defines the Wirtinger presentation - Visualises knot in xy -plane and identifies 'under' crossing as below in z plane. These 'under' crossings can then be identified as 'breakages' to split graph into arcs.
- Prop 2.4 gives full proof with Example 2.5 giving the Wirtinger presentation of a trefoil
- Gives ~~an~~ a method using van Kampens - (not very detailed will research more at a later date)

2/12 Read ~~introduction~~ 'Introductory knot theory' by Martin Christensen chapter 1:

- Covers basics of knot theory again but gives a full breakdown of terminology.

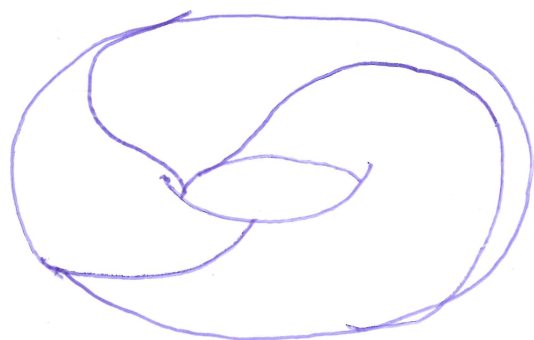
3/12 Extended study of knot group to 2 weeks rather than 1.

Supervisor Signature: ~~Christensen~~

3/12 Des. Torus knot

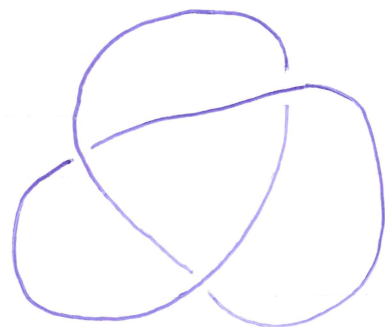
A torus knot is a knot that lies on the surface of an unknotted torus in $\mathbb{R}^3$. Each torus knot is specified by a pair of coprime integers p & q , where the torus knot is trivial if and only if p or q are equal to -1 or 1 .

ex.



where the torus has essentially been 'threaded' through a trefoil knot. This is represented as $(2, -3)$ ~~the trefoil knot~~ torus knot.

where the trefoil knot is the simplest non-trivial knot. The knot has three crossings and can be seen below:



WC
4/12

Week 11

* Meeting Cancelled as not required *

7/12 ^{Log} Read chapter 2.3 of 'Introductory Knot Theory' by Martin Christensen:

- Gives explanation on Van Kampen's thm. (thm. 2.1.2)
- Identifies how knot group identifies isomorphic groups
- Compares 'usefulness' of invariants; knot group and Jones polynomial. Explains how knot group was used to classify torus knots, but is unable to ~~identify~~ ^{distinguish} mirror images

10/12 Need to ask Anitha if an invariant comparison should be included.

10/12 Wrote Introduction for progress report

Supervisor Signature: *Anitha*

W/C
11/12

Week 12

Meeting Plan

~~Report Due~~

- Ask about invariant comparison
- Show current progress report for feedback
- Show current logbook

Log

Logbook-2 Due 15/12

- 11/12 Finished main sections of progress report
- 13/12 Added bibliography
- 14/12 Sorted style with front page of progress report

Supervisor Signature: ~~John Doe~~